

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2024

Bachelor in Computer Applications

Course Title: **DotNet Technology**

Code No: **CACS302** Semester: **V**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is managed code? Give reason to justify the statement "DOT NET framework is platform independent and Language interoperability".
3. What is optional parameter? Write a C# program which stores values in two enumerations, Department and College. It uses two functions to display the data contained in Department and College enumerations.
4. What is difference between value type and reference type in C#? Explain the concept of boxing and unboxing with suitable program.
5. What is interface? Explain with suitable program, how interface can be used to perform multiple inheritance?
6. What are use of finalizer in C#? Explain with suitable program, how base keyword is used to call member function and constructor of parent class?
7. What are keyword used in exception handling? Write a C# program to handle IndexOutOfRangeException and InvalidCastException.
8. What is query syntax in LINQ? Explain the standard query operators "select", "contains", "orderBy" and "where" in LINQ with suitable examples.

Group C

Attempt any TWO questions[2x10=20]

9. a) What is an event? Explain with suitable program, how event is handled in C#.b) What is Lambda expression? Explain the concept of generic delegates with suitable program.
10. How this keyword is used to declare indexer? Write C# program to overload unary (++) and relational (==) operator.
11. What are component of ASP.NET? Explain various validation server control in ASP.NET with suitable program.